

Final Size and Equilibrium Analysis

EdgeBasedModels.jl Vignette 13

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Introduction

The final size of an epidemic is the fraction of the population ultimately infected when the epidemic runs its course ($t \rightarrow \infty$). A remarkable feature of edge-based models is that the final size can be computed without solving any ODEs — it reduces to a fixed-point equation involving the probability generating function (PGF).

The key equation for the final size on a configuration-model network is:

$$\theta_\infty = 1 - T + T \frac{\psi'(\theta_\infty)}{\psi'(1)}$$

where T is the transmissibility (probability of transmission across a single edge before recovery), and $\psi(z)$ is the PGF of the degree distribution. For SIR dynamics with transmission rate β and recovery rate γ , the transmissibility is $T = \beta/(\beta + \gamma)$.

Once θ_∞ is found, the fraction that escapes infection is $\psi(\theta_\infty)$, so the final size is:

$$R_\infty = 1 - \psi(\theta_\infty)$$

For sub-threshold epidemics ($R_0 \leq 1$), the only fixed point is $\theta_\infty = 1$ and $R_\infty = 0$.

This approach follows Newman (2002) and Miller (2007). `EdgeBasedModels.jl` provides `final_size`, `epidemic_probability`, and `confidence_bands` to compute these quantities directly.

Setup

```
using EdgeBasedModels
using OrdinaryDiffEq
using ModelingToolkit
using Plots
```

Basic Final Size Computation

The function `final_size(model)` takes a `StaticConfigurationModel` and returns a named tuple `(R_infinity,  $\theta_\infty$ )` by iterating the fixed-point equation to convergence.

Let us compute the final size for an SIR epidemic on a Poisson network with mean degree $\kappa = 5$, transmission rate $\beta = 1/6 \approx 0.1667$, and recovery rate $\gamma = 0.25$ — anchored to $R_0 = T\kappa = 2$ ($T = \beta/(\beta + \gamma) = 0.4$):

```
pgf = poisson_pgf(5)

sir_prog = DiseaseProgression(
    [DiseaseStage(:I; transmission_rate=1/6), DiseaseStage(:R)],
    [DiseaseTransition(:I, :R, 0.25)];
    entry=:I
)
model = StaticConfigurationModel(pgf, sir_prog)

fs = final_size(model)
println("Final size  $R_\infty$  = $(round(fs.R_infinity * 100, digits=1))%")
println(" $\theta_\infty$  = $(round(fs. $\theta_\infty$ , digits=6))")
```

```
Final size  $R_\infty$  = 79.7%
 $\theta_\infty$  = 0.681275
```

Comparison with ODE Simulation

We can verify the analytic final size by running the ODE model to large t :

```
sir_system = build_sir(pgf, 1/6, 0.25; form=:expanded)
ic = default_initial_conditions(sir_system; ε=1e-2)
prob = ODEProblem(sir_system.system, ic, (0.0, 200.0))
sol = solve(prob, Tsit5(); saveat=0.5)

R_ode = compartment(sol, sir_system, :R)[end]
println("R∞ from ODE at t=200: $(round(R_ode, digits=6))")
println("R∞ from final_size:   $(round(fs.R_infinity, digits=6))")
println("Difference:          $(round(abs(R_ode - fs.R_infinity), digits=8))")
```

```
R∞ from ODE at t=200: 0.800193
R∞ from final_size:   0.796812
Difference:           0.00338075
```

The two values agree to high precision, confirming that `final_size` gives the exact equilibrium without time integration.

Final Size vs R_0

The relationship between R_0 and the final epidemic size produces a classic sigmoid curve. We sweep R_0 by varying the transmission rate β while holding $\gamma = 0.25$ and $\kappa = 5$ fixed (the $R_0 = 2$ canonical anchor sits on this curve at $\beta = 1/6$):

```
γ_val = 0.25
κ_val = 5

β_range = range(0.01, 2.5, length=200)
R0_vals = Float64[]
fs_vals = Float64[]

for β_val in β_range
    T = β_val / (β_val + γ_val)
    R0 = T * κ_val    # Poisson: excess degree = mean degree
    prog = DiseaseProgression(
        [DiseaseStage(:I; transmission_rate=β_val), DiseaseStage(:R)],
        [DiseaseTransition(:I, :R, γ_val)]; entry=:I
    )
```

```

m = StaticConfigurationModel(poisson_pgf( $\kappa_{\text{val}}$ ), prog)
push!(R0_vals, R0)
push!(fs_vals, final_size(m).R_infinity)
end

plot(R0_vals, fs_vals, lw=2, color=:steelblue, label="Final size  $R_{\infty}$ ")
vline!([1.0], linestyle=:dash, color=:red, label=" $R_0 = 1$ ")
xlabel!(" $R_0$ ")
ylabel!("Final size  $R_{\infty}$ ")
title!("Final Size vs  $R_0$  (Poisson network)")

```

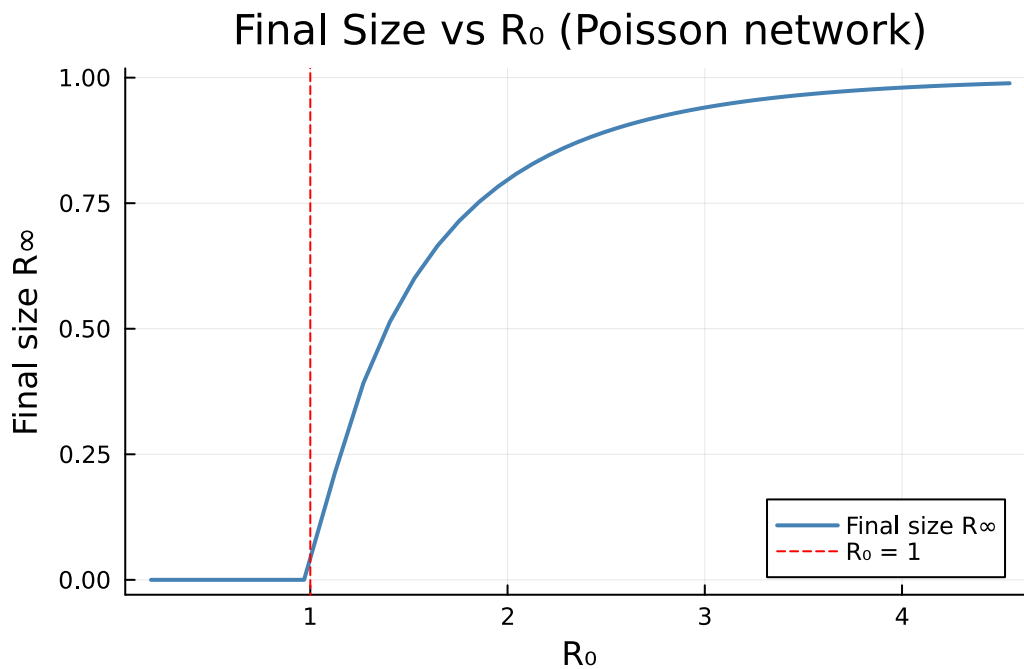


Figure 1: Final epidemic size as a function of R_0 for a Poisson(5) network. The epidemic threshold at $R_0 = 1$ is marked by a dashed line.

Key observations:

- The final size is zero for $R_0 \leq 1$ (no epidemic).
- Just above the threshold the final size grows continuously from zero — a transcritical bifurcation.
- For large R_0 the final size approaches 1; nearly everyone is infected.

Effect of Degree Distribution

Network heterogeneity has a strong effect on the final size. For a given mean degree, a more heterogeneous degree distribution produces a larger epidemic because high-degree nodes act as super-spreaders.

We compare a Poisson network (low variance, $\text{Var}(k) = \kappa$) with a heterogeneous two-point distribution $P(k = 0) = 1 - \kappa/k_{\max}$, $P(k = k_{\max}) = \kappa/k_{\max}$ (high variance) at the same mean degree $\kappa = 4$:

```
κ_cmp = 4
k_max = 12

# Two-point distribution: P(0) = 1 - κ/k_max, P(k_max) = κ/k_max
probs_twopoint = zeros(k_max + 1)
probs_twopoint[1] = 1.0 - κ_cmp / k_max
probs_twopoint[k_max + 1] = κ_cmp / k_max
pgf_twopoint = polynomial_pgf(probs_twopoint)

pgf_poisson = poisson_pgf(κ_cmp)

T_range = range(0.01, 0.99, length=200)
fs_poisson = Float64[]
fs_hetero = Float64[]

for T_val in T_range
    # Back out β from T = β/(β+γ) → β = T·γ/(1-T)
    γ_fix = 0.25
    β_val = T_val * γ_fix / (1 - T_val)

    prog = DiseaseProgression(
        [DiseaseStage(:I; transmission_rate=β_val), DiseaseStage(:R)],
        [DiseaseTransition(:I, :R, γ_fix)]; entry=:I
    )
    m_p = StaticConfigurationModel(pgf_poisson, prog)
    m_h = StaticConfigurationModel(pgf_twopoint, prog)
    push!(fs_poisson, final_size(m_p).R_infinity)
    push!(fs_hetero, final_size(m_h).R_infinity)
end

plot(T_range, fs_poisson, lw=2, label="Poisson(κ=$κ_cmp)", color=:steelblue)
plot!(T_range, fs_hetero, lw=2, label="Two-point(0,$k_max)", color=:darkorange)
xlabel!("Transmissibility T")
```

```
ylabel!("Final size  $R_\infty$ ")
title!("Effect of Degree Heterogeneity on Final Size")
```

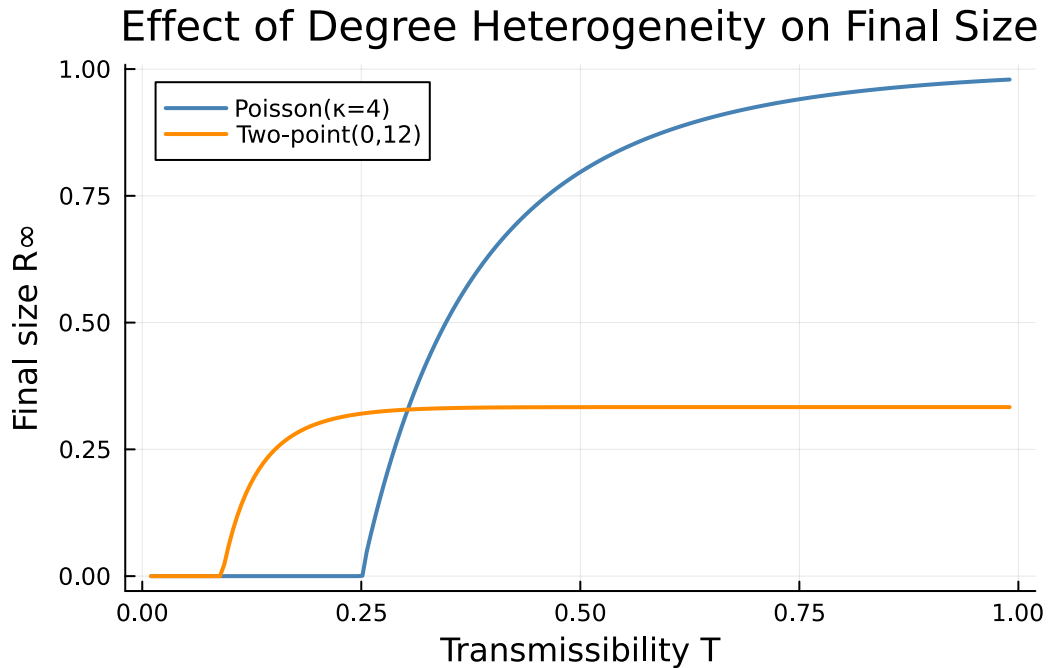


Figure 2: Final size vs transmissibility T for Poisson and heterogeneous (two-point) networks with the same mean degree $\kappa = 4$. The heterogeneous network has a larger final size and a lower epidemic threshold.

The heterogeneous network has:

- A lower epidemic threshold (smaller critical T), because the excess degree ratio $\psi''(1)/\psi'(1)$ is larger when variance is higher.
- A larger final size at any given transmissibility above threshold, because high-degree nodes amplify spread.

Epidemic Probability

The `epidemic_probability(model)` function computes the probability that a single initial infection triggers a major epidemic (as opposed to dying out stochastically in the early phase). This is based on the extinction probability q of the branching process approximation, satisfying:

$$q = \frac{\psi'(1 - T + Tq)}{\psi'(1)}$$

The epidemic probability is then $1 - \psi(1 - T + Tq)$. For $R_0 \leq 1$, the epidemic probability is zero.

```

γ_ep = 0.25
κ_ep = 5

β_ep_range = range(0.01, 2.5, length=200)
R0_ep = Float64[]
fs_ep = Float64[]
ep_ep = Float64[]

for β_val in β_ep_range
    T = β_val / (β_val + γ_ep)
    R0 = T * κ_ep
    prog = DiseaseProgression(
        [DiseaseStage(:I; transmission_rate=β_val), DiseaseStage(:R)],
        [DiseaseTransition(:I, :R, γ_ep)]; entry=:I
    )
    m = StaticConfigurationModel(poisson_pgf(κ_ep), prog)
    push!(R0_ep, R0)
    push!(fs_ep, final_size(m).R_infinity)
    push!(ep_ep, epidemic_probability(m))
end

plot(R0_ep, fs_ep, lw=2, label="Final size R∞", color=:steelblue)
plot!(R0_ep, ep_ep, lw=2, label="Epidemic probability", color=:darkorange, linestyle=:dash)
vline!([1.0], linestyle=:dot, color=:red, label="R₀ = 1")
xlabel!("R₀")
ylabel!("Probability / Fraction")
title!("Final Size vs Epidemic Probability")

```

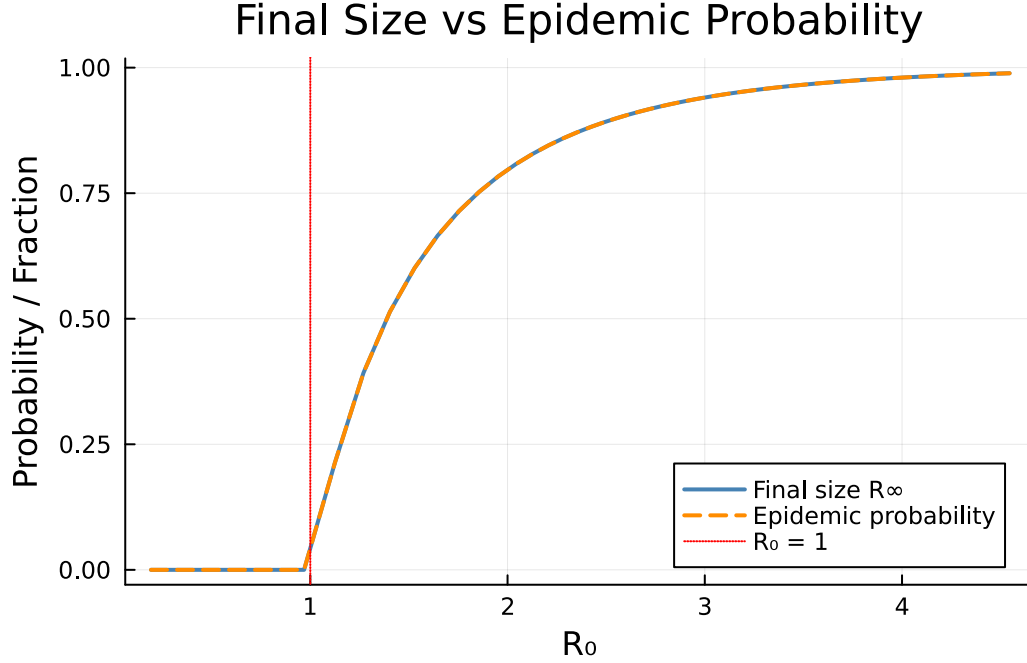


Figure 3: Final size and epidemic probability as functions of R_0 . The epidemic probability is always smaller than the final size: even when a major epidemic occurs, some individuals escape infection.

Both quantities are zero below threshold and increase with R_0 , but they are not the same. The epidemic probability gives the chance of a large outbreak *occurring*; the final size gives the fraction infected *given that it does*. The epidemic probability is always smaller than the final size.

Confidence Bands for Finite Populations

The analytic final size is an asymptotic result valid for infinite populations. For finite populations of size N , the actual final size is a random variable with variance that decreases as N grows. The function `confidence_bands(model, N)` computes CLT-based confidence intervals using the asymptotic variance formula from Ball (2021):

$$\sigma^2 = R_\infty(1 - R_\infty) + \frac{R_\infty \cdot T \cdot \psi'(\theta_\infty)(1 - \theta_\infty)}{\psi'(1)(1 - h'(\theta_\infty))^2}$$

where $h'(\theta) = T \cdot \psi''(\theta)/\psi'(1)$ is the derivative of the fixed-point map. The standard error scales as $\sigma/\sqrt{N}$.


```

pgf_cb = poisson_pgf(5)
prog_cb = DiseaseProgression(
    [DiseaseStage(:I; transmission_rate=1/6), DiseaseStage(:R)],
    [DiseaseTransition(:I, :R, 0.25)]; entry=:I
)
model_cb = StaticConfigurationModel(pgf_cb, prog_cb)

for N in [100, 1_000, 10_000]
    cb = confidence_bands(model_cb, N)
    println("N = $(lpad(N, 6)): R $\infty$  = $(round(cb.mean, digits=4)) " *
        "95% CI = [$(round(cb.lower, digits=4)), $(round(cb.upper, digits=4))] " *
        "SE = $(round(cb.std_error, digits=5))")
end

```

```

N =    100: R $\infty$  = 0.7968  95% CI = [0.7048, 0.8889]  SE = 0.04695
N =   1000: R $\infty$  = 0.7968  95% CI = [0.7677, 0.8259]  SE = 0.01485
N =  10000: R $\infty$  = 0.7968  95% CI = [0.7876, 0.806]   SE = 0.0047

```

```

 $\gamma$ _cb = 0.25
 $\kappa$ _cb = 5

 $\beta$ _cb_range = range(0.025, 2.5, length=150)
R0_cb = Float64[]
mean_cb = Float64[]
lower_100 = Float64[]
upper_100 = Float64[]
lower_1k = Float64[]
upper_1k = Float64[]
lower_10k = Float64[]
upper_10k = Float64[]

for  $\beta$ _val in  $\beta$ _cb_range
    T =  $\beta$ _val / ( $\beta$ _val +  $\gamma$ _cb)
    R0 = T *  $\kappa$ _cb
    prog = DiseaseProgression(
        [DiseaseStage(:I; transmission_rate= $\beta$ _val), DiseaseStage(:R)],
        [DiseaseTransition(:I, :R,  $\gamma$ _cb)]; entry=:I
    )
    m = StaticConfigurationModel(poisson_pgf( $\kappa$ _cb), prog)

    cb100 = confidence_bands(m, 100)

```

```

cb1k = confidence_bands(m, 1_000)
cb10k = confidence_bands(m, 10_000)

push!(R0_cb, R0)
push!(mean_cb, cb1k.mean)
push!(lower_100, cb100.lower); push!(upper_100, cb100.upper)
push!(lower_1k, cb1k.lower); push!(upper_1k, cb1k.upper)
push!(lower_10k, cb10k.lower); push!(upper_10k, cb10k.upper)
end

plot(R0_cb, mean_cb, lw=2, color=:black, label="Mean  $R_\infty$ ")
plot!(R0_cb, lower_100, fillrange=upper_100, alpha=0.15, color=:red, label="N = 100")
plot!(R0_cb, lower_1k, fillrange=upper_1k, alpha=0.20, color=:steelblue, label="N = 1,000")
plot!(R0_cb, lower_10k, fillrange=upper_10k, alpha=0.25, color=:seagreen, label="N = 10,000")
vline!([1.0], linestyle=:dot, color=:grey, label="R0 = 1")
xlabel!("R0")
ylabel!("Final size  $R_\infty$ ")
title!("Confidence Bands by Population Size")

```

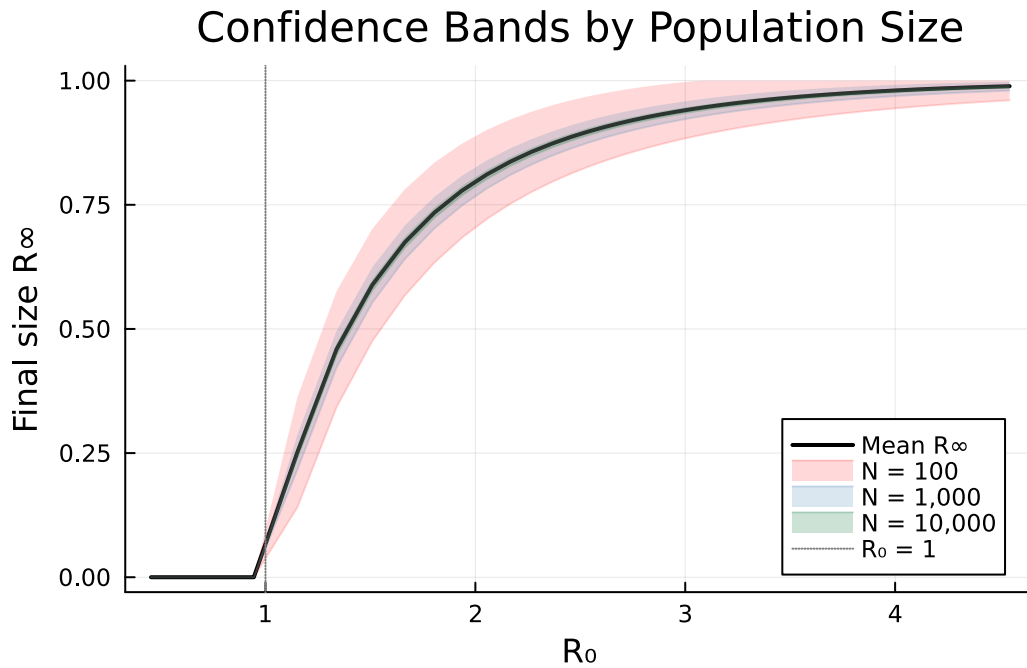


Figure 4: 95% confidence bands for the final size as a function of R_0 , for population sizes $N = 100$, 1000, and 10000. Bands narrow with increasing N .

As expected, the confidence intervals narrow with increasing N , converging to the deterministic final size in the infinite-population limit.

Final Size Sensitivity to Network Connectivity

We can explore how the final size depends jointly on the mean degree κ and the transmissibility $T = \beta/(\beta + \gamma)$. This produces a contour plot that reveals the epidemic threshold curve.

```
κ_range_2d = range(0.5, 10.0, length=100)
T_range_2d = range(0.01, 0.99, length=100)

fs_matrix = zeros(length(κ_range_2d), length(T_range_2d))

for (i, κ_val) in enumerate(κ_range_2d)
  for (j, T_val) in enumerate(T_range_2d)
    γ_fix = 0.25
    β_val = T_val * γ_fix / (1 - T_val)
    prog = DiseaseProgression(
      [DiseaseStage(:I; transmission_rate=β_val), DiseaseStage(:R)],
      [DiseaseTransition(:I, :R, γ_fix)]; entry=:I
    )
    m = StaticConfigurationModel(poisson_pgf(κ_val), prog)
    fs_matrix[i, j] = final_size(m).R_infinity
  end
end

heatmap(collect(κ_range_2d), collect(T_range_2d), fs_matrix',
  xlabel="Mean degree κ",
  ylabel="Transmissibility T",
  title="Final Size R∞",
  colorbar_title="R∞",
  color=:viridis)
contour!(collect(κ_range_2d), collect(T_range_2d), fs_matrix',
  levels=[0.01], linewidth=2, linecolor=:white, label="")
```

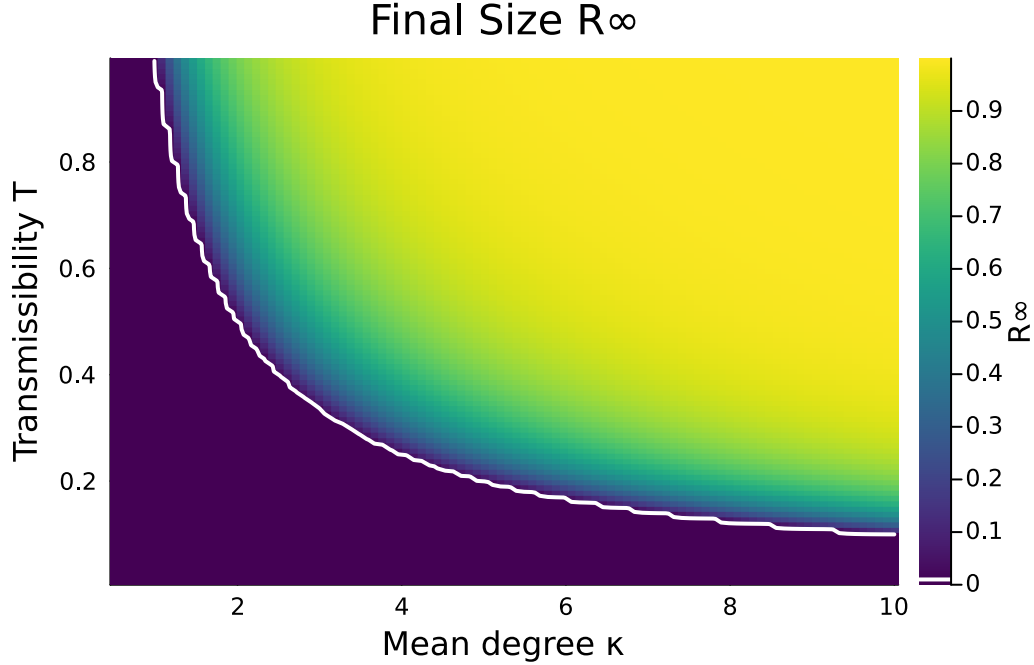


Figure 5: Final size as a function of mean degree κ and transmissibility T . The white contour marks $R_0 = 0.01$ (approximate epidemic threshold). The epidemic region lies above and to the right.

The epidemic threshold follows the curve $T \cdot \kappa = 1$ (i.e., $R_0 = 1$) for Poisson networks, since the excess degree ratio equals the mean degree. Below this curve no epidemic occurs; above it the final size increases smoothly.

Summary

EdgeBasedModels.jl provides efficient analytic tools for epidemic equilibrium analysis:

- `final_size(model)` computes the exact final epidemic size R_∞ via fixed-point iteration — orders of magnitude faster than solving ODEs to convergence.
- `epidemic_probability(model)` gives the probability that a single introduction triggers a major outbreak, based on the branching-process extinction probability.
- `confidence_bands(model, N)` provides CLT-based confidence intervals for the final size in finite populations, following Ball (2021).
- Degree heterogeneity amplifies both R_0 and the final size: networks with higher degree variance have lower epidemic thresholds and larger outbreaks.
- The contour plot of final size in the (κ, T) plane provides a clear picture of the epidemic threshold and its dependence on network connectivity.

These analytic results complement the ODE simulations in earlier vignettes and are particularly valuable for parameter space exploration, sensitivity analysis, and rapid model comparison.

Simulation validation

We compare the analytic `final_size` curve against direct stochastic simulation. For each $R_0 \in \{0.5, 0.75, \dots, 4.0\}$ we hold $\gamma = 0.25, \kappa = 5$ fixed and back out $\beta = R_0\gamma/(\kappa - R_0)$. We then build an Erdős–Rényi graph ($N = 2000, p = \kappa/(N - 1)$) for each replicate, run a `DirectSSA` trajectory to $t = 200$, and report the empirical final size as a 95% band across replicates and graphs.

```
include("../_validation.jl")

R0_sweep = collect(0.5:0.25:4.0)
γ_v = 0.25
κ_v = 5.0
N_v = 2000

prog_factory = function (R0)
    β_v = R0 * γ_v / κ_v
    DiseaseProgression(
        [DiseaseStage(:I; transmission_rate = β_v), DiseaseStage(:R)],
        [DiseaseTransition(:I, :R, γ_v)]; entry = :I)
end

params_factory = function (R0)
    β_v = R0 * γ_v / κ_v
    Dict(:β ⇒ β_v, :γ ⇒ γ_v)
end

sweep = gillespie_final_size_sweep(
    prog_factory, params_factory,
    poisson_graph_builder(N_v, κ_v),
    R0_sweep;
    N = N_v, n_graphs = 3, nsims_per_graph = 12,
    tspan = (0.0, 200.0), seed_fraction = 0.01, recovered = :R)
```

(values = [0.5, 0.75, 1.0, 1.25, 1.5, 1.75, 2.0, 2.25, 2.5, 2.75, 3.0, 3.25, 3.5, 3.75, 4.0], me

```
fs_curve = Float64[]
for R0 in R0_sweep
    β_v = R0 * γ_v / κ_v
    prog = DiseaseProgression(
```

```

    [DiseaseStage(:I; transmission_rate =  $\beta_v$ ), DiseaseStage(:R)],
    [DiseaseTransition(:I, :R,  $\gamma_v$ )]; entry = :I)
m = StaticConfigurationModel(poisson_pgf( $\kappa_v$ ), prog)
push!(fs_curve, EdgeBasedModels.final_size(m).R_infinity)
end

# Conditioned on a major outbreak (drop final sizes < 5%):
ssa_major_mean = [(let v = filter( $\geq$ (0.05), s); isempty(v) ? 0.0 : mean(v) end)
    for s in sweep.all]

plot(R0_sweep, fs_curve, lw = 2, color = :black,
    label = "Analytic  $R_\infty$ ")
plot!(sweep.values, sweep.means,
    ribbon = (sweep.means .- sweep.q025, sweep.q975 .- sweep.means),
    color = :steelblue, fillalpha = 0.25, linealpha = 0.6, lw = 1,
    label = "SSA mean (95% band)")
plot!(sweep.values, ssa_major_mean, lw = 1.5, ls = :dash,
    color = :grey40, label = "SSA major-outbreak mean ( $R_\infty \geq 0.05$ )")
vline!([1.0], ls = :dot, color = :red, label = " $R_0 = 1$ ")
xlabel!(" $R_0$ ")
ylabel!("Final size  $R_\infty$ ")
title!("Analytic vs simulated final size on Poisson( $\kappa=5$ ) network")

```

analytic vs simulated final size on Poisson($\kappa=5$) net

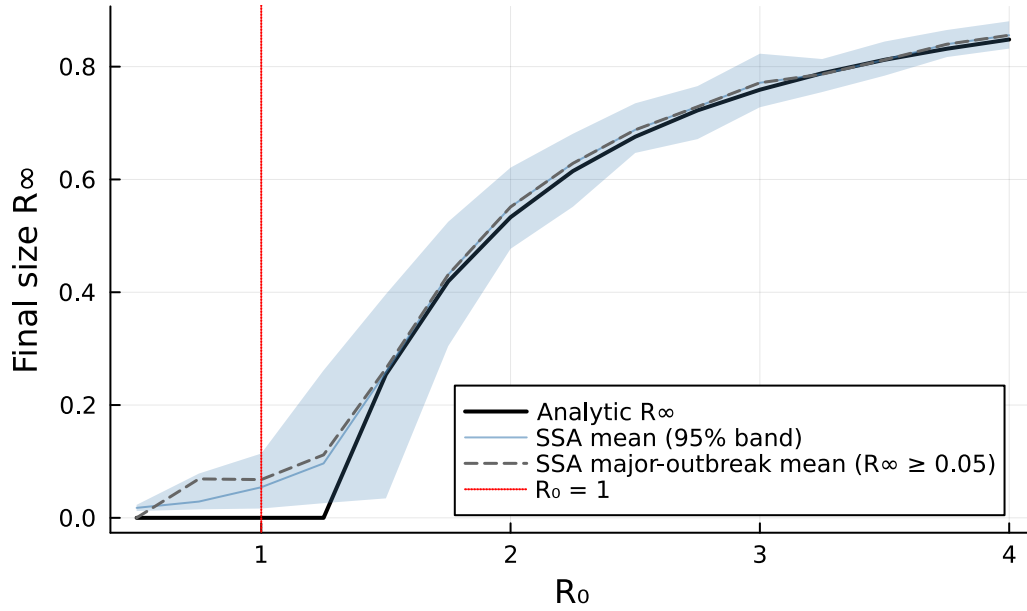


Figure 6: Analytic final size (line) versus DirectSSA on Erdős–Rényi graphs ($N=2000$, ribbon = 95% empirical interval across 36 trajectories per R_0). Major / minor outbreak bimodality near the threshold widens the band; subset-conditioned curve overlaid in grey.

The analytic curve and the major-outbreak conditional mean agree closely across the full sweep. The unconditioned mean (with 95% band) sags below the deterministic curve near the threshold because some trajectories die out stochastically — this is the regime in which `epidemic_probability` is materially less than 1 (cf. the previous figure). For $R_0 \gtrsim 2$ the epidemic probability is essentially 1 and the two curves coincide.